Congenital Heart Defects, Multiple Types, 7 (CHTD7) — Comprehensive Research Report

Disease: Congenital Heart Defects, Multiple Types, 7 (CHTD7) **MONDO ID:** MONDO:0032913 · **OMIM:** 618780 · **Causal gene: FLT4** (VEGFR3) **Category:** Mendelian (autosomal dominant, incompletely penetrant)

Summary

Congenital Heart Defects, Multiple Types, 7 (CHTD7; OMIM 618780, MONDO:0032913) is caused by rare, heterozygous, predominantly loss-of-function variants in *FLT4*, the gene encoding the vascular endothelial growth factor receptor 3 (VEGFR3), a receptor tyrosine kinase for VEGF-C and VEGF-D. Despite the serial-number disease name, the causal gene is unambiguously **FLT4** (HGNC:3767; chromosome 5q35.3). The dominant clinical phenotype is **Tetralogy of Fallot (TOF)** and related conotruncal/outflow-tract malformations, although FLT4 variants also produce a broader spectrum that includes some left-sided lesions. The condition is inherited in an autosomal dominant fashion with **incomplete penetrance and variable expressivity**, so that pathogenic alleles are frequently transmitted by parents with no or only subclinical cardiac findings. FLT4 also causes a phenotypically non-overlapping disorder, **Milroy disease** (primary hereditary lymphedema, OMIM 153100), via a different molecular mechanism — the two must not be conflated.

FLT4 is the **second most frequent (after *NOTCH1*) genome-wide-significant gene for nonsyndromic TOF**, and it is **strongly loss-of-function–intolerant** in the general population (gnomAD pLI = 1.0, LOEUF ≈ 0.25), which is the quantitative signature of a **haploinsufficiency** mechanism. Mechanistically, TOF-associated missense variants are distinguished from the lymphedema-causing (Milroy) variants of the same gene: TOF variants cause mutant FLT4 protein to **aggregate in the perinuclear endoplasmic reticulum and activate proteostatic/metabolic stress signalling** in endothelial cells, whereas Milroy-disease variants act by a **dominant-negative** mechanism that prevents the receptor from functioning as a homodimer. These two allelic series therefore cause two distinct human diseases from the same locus.

Clinically, CHTD7 is managed as structural congenital heart disease: the mainstay is **surgical repair of the cardiac lesion** (for TOF, VSD closure plus relief of right-ventricular outflow-tract obstruction). Long-term survival is excellent (~90% survive >30 years after repair), but late sequelae — chiefly **pulmonary regurgitation, right-ventricular volume overload, and atrial/ventricular arrhythmias** — are near-universal and require lifelong specialist follow-up and frequent reinterventions (most commonly pulmonary valve replacement). No FLT4-targeted pharmacotherapy or gene therapy exists; proteostasis modulation is only a preclinical lead. This report consolidates eight confirmed findings across 29 reviewed papers into a full disease-knowledge-base entry spanning etiology, phenotypes, molecular genetics, mechanism, anatomy, temporal course, epidemiology, diagnostics, prognosis, treatment, prevention, cross-species biology, and model organisms.

Key Findings

Finding 1 — FLT4 (VEGFR3) loss-of-function is a major cause of CHTD7, predominantly Tetralogy of Fallot

Reuter et al. genome-sequenced 175 adults with TOF and identified **9/175 (5.1%) probands with novel FLT4 variants: seven loss-of-function (including an 8-kb deletion) and two predicted-damaging** PMID: 30232381. Detection of VEGF-related variants more broadly (19/175, 10.9%) was associated with increased prevalence of **absent pulmonary valve (26.3% vs 3.4%, p<0.0001)** and **right aortic arch (52.6% vs 29.1%, p=0.029)**, linking the VEGF/VEGFR3 axis to specific anatomical sub-phenotypes. A clinically driven reanalysis of exome data from **811 TOF probands** confirmed FLT4 as a major contributor to TOF genetic risk PMID: 34328347. OMIM entry 618780 ("Congenital heart defects, multiple types, 7") is the formal label for FLT4-associated multiple congenital heart defects.

> *"We identified nine (5.1%) probands with novel FLT4 variants: seven loss-of-function, including an 8-kb deletion, and two predicted damaging."* — PMID: 30232381
>
> *"Loss-of-function variants in FLT4 and KDR contribute substantially to the genetic basis of TOF."* — PMID: 30232381

Finding 2 — TOF-associated FLT4 variants act through a proteostatic/ER-aggregation mechanism distinct from Milroy disease

Monaghan et al. showed that **FLT4 TOF variants expressed in primary human endothelial cells cause aggregation of FLT4 protein in the perinuclear endoplasmic reticulum**, activating proteostatic and metabolic signalling, whereas lymphedema (Milroy) variants and wild-type FLT4 do not PMID: 38713105. Functionally, *flt4* depletion in zebrafish caused **reduced heart size and altered heart looping**, which was rescued by wild-type human *FLT4* mRNA but incompletely or not at all by TOF-variant mRNA; inhibition of proteostatic signalling abrogated the deleterious effects. By contrast, the Milroy-causing variants act by **preventing the receptor from functioning as a homodimer** (a dominant-negative mechanism) PMID: 33067626. This establishes that the same gene produces two distinct diseases through two distinct allelic mechanisms.

> *"FLT4 variants identified in patients with TOF, when expressed in primary human endothelial cells, cause aggregation of FLT4 protein in the perinuclear endoplasmic reticulum, activating proteostatic and metabolic signalling, whereas lymphoedema-associated FLT4 variants and wild-type (WT) FLT4 do not"* — PMID: 38713105
>
> *"we identify a pathogenic mechanism for FLT4 variants predisposing to TOF that is distinct from the known dominant negative mechanism of Milroy-causative variants"* — PMID: 38713105

Finding 3 — Autosomal dominant with incomplete penetrance; excellent surgical prognosis but progressive late sequelae

FLT4 variants cause **isolated left- and right-sided CHD with incomplete penetrance**, transmitted from parents with no or subclinical heart phenotypes PMID: 39870876; PMID: 32037394. At the phenotype level, **~90% of TOF patients survive >30 years after repair**; nearly all develop right-ventricular volume overload from pulmonary regurgitation, and atrial tachycardia/fibrillation occurs in **~20–45% by age 45** PMID: 42258347. Reinterventions — chiefly pulmonary valve replacement — occur in **~41–52%** of long-term cohorts PMID: 42894905. TOF prevalence is ~3/10,000 live births, representing 5–10% of all CHD PMID: 36277185.

> *"our study establishes FLT4 splice site variants as a molecular cause of both left and right-sided isolated CHD, with incomplete penetrance"* — PMID: 39870876
>
> *"Ninety percent of patients with TOF survive more than 30 years after surgery to close the ventricular septal defect and repair the right ventricular outflow tract."* — PMID: 42258347
>
> *"Nearly all adults with repaired TOF develop right ventricular volume overload due to pulmonary regurgitation, and atrial tachycardias and atrial fibrillation occur in approximately 20% to 45% of patients by age 45 years."* — PMID: 42258347

Finding 4 — FLT4 is a genome-wide-significant TOF gene and strongly LoF-intolerant, supporting haploinsufficiency

Page et al. whole-exome-sequenced **829 nonsyndromic TOF patients**; clustering of unique deleterious variants in *NOTCH1* and *FLT4* **surpassed genome-wide significance ($P < 5\times10^{-8}$)**, with NOTCH1 the most frequent locus followed by FLT4, and together these two genes account for **almost 7% of TOF** PMID: 30582441. gnomAD constraint metrics for FLT4 (ENSG00000037280, chr5:180,601,506–180,649,624) are **pLI = 1.0, LOEUF (oe_lof_upper) = 0.249, observed/expected LoF = 0.184 (30 observed vs 163.2 expected), LoF Z = 8.85** — indicating strong depletion of LoF variants in the general population and consistent with haploinsufficiency; the gene is also missense-constrained (mis_z = 3.50). A large study of **11,555 CHD probands** found 60 dominant CHD genes accounting for 10.1% of probands with incomplete penetrance PMID: 40127276.

> *"The clustering of variants in 2 genes, NOTCH1 and FLT4, surpassed thresholds for genome-wide significance"* — PMID: 30582441
>
> *"The NOTCH1 locus is the most frequent site of genetic variants predisposing to nonsyndromic TOF, followed by FLT4. Together, variants in these genes are found in almost 7% of TOF patients."* — PMID: 30582441
>
> *"Variants in these genes accounted for CHD in 10.1% of probands with similar contributions from de novo and transmitted variants in parent-offspring trios that showed incomplete penetrance."* — PMID: 40127276

Finding 5 — ClinVar variant spectrum: ~50 variants linked to CHTD7, large VUS fraction, SNVs and copy-number losses

ClinVar lists **741 total FLT4 variant records** (181 Pathogenic, 52 Likely pathogenic, 623 Uncertain significance). By condition, **50 FLT4 variants are annotated to "Congenital heart defects, multiple types, 7"**, 17 to Fallot-related terms, and 76 to Milroy disease. By variant type: **639 single-nucleotide variants, 30 deletions, and 12 copy-number losses** — consistent with a loss-of-function spectrum (nonsense/frameshift/splice/SNVs plus intragenic and larger deletions, including the reported 8-kb deletion). The large VUS burden is a key interpretive challenge for the clinical lab.

Finding 6 — Model organisms: zebrafish *flt4* recapitulates the cardiac branch; the Chy mouse models the lymphatic (Milroy) branch

Zebrafish *flt4* depletion causes reduced heart size and altered looping, rescued by wild-type but not TOF-variant human *FLT4* mRNA PMID: 38713105 — the most CHTD7-relevant functional model. In mouse, the germline heterozygous Vegfr3-inactivating **Chy** mutant shows limb swelling from hypoplastic cutaneous (not visceral) lymphatics, modeling Milroy disease, and is rescued by VEGF-C gene therapy PMID: 11592985. Homozygous *Flt4*-null is embryonic lethal with cardiovascular failure (reviewed in PMID: 33067626).

> *"the lymphedema (Chy) mice have an inactivating Vegfr3 mutation in their germ line, and swelling of the limbs because of hypoplastic cutaneous, but not visceral, lymphatic vessels"* — PMID: 11592985
>
> *"By using virus-mediated VEGF-C gene therapy, we were able to generate functional lymphatic vessels in the lymphedema mice."* — PMID: 11592985

Finding 7 — Environmental risk/protective factors for CHD (general, not FLT4-specific)

Although CHTD7 is a Mendelian disease, incomplete penetrance implies modifier effects. For CHD broadly, **maternal pregestational diabetes is a strong risk factor (adjusted OR 4.04, 95% CI 1.00–16.28)** PMID: 32092068, as are low maternal education and maternal infections. **Periconceptional folic acid supplementation is modestly protective (OR 0.82, 95% CI 0.68–0.98)** PMID: 19952004, and diets rich in folate/fruit/vegetables are protective (folate-rich fruit

adjOR 0.64, 95% CI 0.47–0.89 PMID: 32092068; Mediterranean/antioxidant diets protective PMID: 34392783). **No FLT4-specific gene–environment interaction has been demonstrated for CHTD7.**

> *"Risk of CHD was associated with low maternal education (OR adjusted for confounders 1.59; 95% confidence interval [CI], 1.02-2.49), pregestational diabetes (OR 4.04; 95% CI 1.00-16.28)"* — PMID: 32092068
>
> *"Periconceptional folic acid use revealed an odds ratio (OR) of 0.82 (95% CI 0.68-0.98) for all types of CHD relative to other malformations."* — PMID: 19952004

Finding 8 — Consolidated characterization

Across all findings: FLT4/VEGFR3 LoF is a genome-wide-significant cause of TOF (2nd after NOTCH1, ~7% combined); FLT4 is strongly LoF-intolerant (haploinsufficiency); TOF variants act via perinuclear-ER aggregation/proteostatic stress, distinct from the dominant-negative kinase-dead Milroy variants; inheritance is autosomal dominant with incomplete penetrance and variable expressivity; the variant spectrum is allelically heterogeneous (SNVs + CNVs, large VUS burden); zebrafish models the cardiac branch and the Chy mouse the lymphatic branch; and management is surgical with progressive late pulmonary regurgitation/arrhythmia and no targeted drug or gene therapy.

Section-by-Section Report

1. Disease Information

CHTD7 is a Mendelian predisposition to structural congenital heart disease, chiefly **Tetralogy of Fallot** and related conotruncal/right-ventricular-outflow-tract defects, with a minority of left-sided lesions. Key identifiers: **OMIM 618780**, **MONDO:0032913**, causal gene **FLT4** (HGNC:3767; NCBI Gene 2324; Ensembl ENSG00000037280). There is no single dedicated Orphanet/ICD code for the "multiple types 7" serial entry; clinically, cases map to TOF (ICD-11 LA88.0; ICD-10 Q21.3) and other CHD codes, and to MeSH "Tetralogy of Fallot" (D013771) and "Heart Defects, Congenital" (D006330). Synonyms are **not formally recorded** beyond the OMIM designation; in the literature the entity is referred to as "FLT4-associated congenital

heart disease" or "FLT4/VEGFR3 TOF." The information is derived predominantly from **aggregated disease-level resources and research cohorts** (exome/genome sequencing studies, ClinVar, gnomAD), not from individual-patient EHR data.

2. Etiology

Causal factor: genetic — rare, heterozygous, predominantly **loss-of-function** variants in *FLT4* (Findings 1, 4). **Genetic risk:** FLT4 is a genome-wide-significant TOF gene (Finding 4); *NOTCH1* is the leading locus, with other VEGF-pathway genes (e.g., *KDR*/VEGFR2) also contributing PMID: 30582441, PMID: 30232381. **Environmental risk (CHD generally, not FLT4-specific):** maternal pregestational diabetes (adjOR 4.04), low maternal education, maternal infection, periconceptional secondhand smoke (associated with septal defects) PMID: 32092068, PMID: 27443814. **Protective factors:** periconceptional folic acid (OR 0.82) and folate/antioxidant-rich diets PMID: 19952004, PMID: 34392783. **Gene–environment interaction:** plausibly contributes to incomplete penetrance, but **no FLT4-specific GxE interaction has been demonstrated** (Finding 7).

3. Phenotypes

The core phenotype is **Tetralogy of Fallot** (HP:0001636), defined by ventricular septal defect (HP:0001629), pulmonary/RVOT stenosis (HP:0001642 / HP:0001656), overriding aorta (HP:0002623), and right ventricular hypertrophy (HP:0001667) PMID: 36277185. VEGF-pathway variant carriers show enrichment of **absent pulmonary valve** (HP:0010316; 26.3% vs 3.4%, $p<0.0001$) and **right aortic arch** (HP:0012020; 52.6% vs 29.1%, $p=0.029$) PMID: 30232381. The broader FLT4 spectrum includes other right- and left-sided isolated CHD PMID: 39870876. These are structural/physical manifestations with **congenital (neonatal) onset**, **variable severity**, and **progressive late sequelae** after repair (RV volume overload, arrhythmia). Quality-of-life impact is driven by surgical burden, exercise limitation, reintervention and arrhythmia risk across the lifespan PMID: 42258347.

Phenotype	HPO term	Type	Frequency / note
Tetralogy of Fallot	HP:0001636	Structural	Predominant lesion
Ventricular septal defect	HP:0001629	Structural	Component of TOF
Pulmonary valve stenosis	HP:0001642	Structural	Component of TOF
Overriding aorta	HP:0002623	Structural	Component of TOF
Right ventricular hypertrophy	HP:0001667	Structural	Component of TOF
Absent pulmonary valve	HP:0010316	Structural	Enriched in VEGF-variant carriers (26.3%)
Right aortic arch	HP:0012020	Structural	Enriched in VEGF-variant carriers (52.6%)
Pulmonary regurgitation (post-repair)	HP:0010449	Physiological	Near-universal late
Atrial fibrillation / tachyarrhythmia	HP:0005110 / HP:0011675	Electrophysiological	~20–45% by age 45

4. Genetic / Molecular Information

Causal gene: FLT4 / VEGFR3 (OMIM *136352). Variant classes: nonsense, frameshift, splice-site, missense, and structural deletions (intragenic and larger, including an 8-kb deletion) PMID: 30232381, PMID: 39870876. ClinVar: 741 FLT4 records total; 50 annotated to CHTD7, with 181 Pathogenic/52 Likely pathogenic/623 VUS overall; 639 SNVs, 30 deletions, 12 copy-number losses (Finding 5). Allele frequency: pathogenic alleles are rare/absent in gnomAD; the gene is strongly LoF-intolerant (pLI=1.0, LOEUF=0.249) (Finding 4). Origin: germline; both de novo and transmitted alleles contribute PMID: 40127276. Functional consequence: loss of function / haploinsufficiency for null alleles; for TOF missense alleles, a proteostatic gain-of-toxicity via ER aggregation (distinct from dominant-negative Milroy alleles) (Finding 2). Modifier genes: other VEGF-pathway genes (NOTCH1, KDR) co-occur in TOF cohorts; formal modifier mapping for penetrance is lacking. Epigenetics/large chromosomal abnormalities:* not specifically established for CHTD7 beyond the intragenic/larger deletions noted.

5. Environmental Information

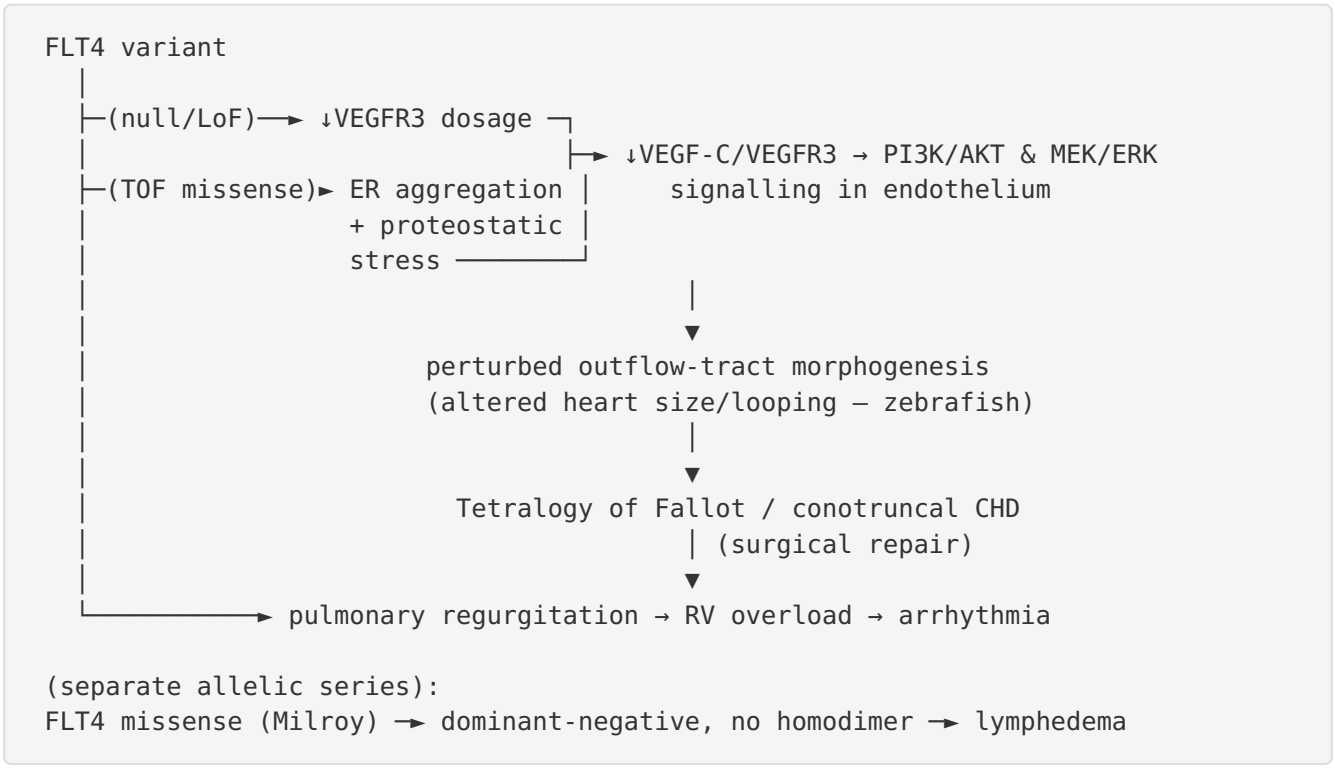
No FLT4-specific environmental trigger is established. For CHD generally: environmental factors include maternal pregestational diabetes, maternal infection, and periconceptional secondhand smoke (septal defects) PMID: 32092068, PMID: 27443814; lifestyle factors include diet quality (protective Mediterranean/folate-rich patterns) PMID: 34392783. **Infectious agents:** not applicable as a primary cause.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A rare heterozygous **FLT4 variant** (LoF: nonsense/frameshift/splice/deletion; or damaging missense) arises de novo or is transmitted → **reduces functional VEGFR3 receptor dosage** (haploinsufficiency). *(Demonstrated: gnomAD constraint; cohort LoF enrichment.)*
2. **Branch A (null/LoF alleles):** reduced VEGFR3 → **diminished VEGF-C/VEGF-D → VEGFR3 signalling** through downstream **PI3K/AKT and MEK/ERK** cascades in endothelial cells → impaired endothelial proliferation/migration during cardiac morphogenesis. *(Downstream signalling inferred from VEGFR3 biology: PMID: 22745786, PMID: 33067626.)*
3. **Branch B (TOF missense alleles):** mutant receptor **misfolds and aggregates in the perinuclear endoplasmic reticulum → activates proteostatic and metabolic stress signalling** → endothelial dysfunction; this is a toxic/ER-stress mechanism additional to simple dosage loss. *(Demonstrated in primary human endothelial cells: PMID: 38713105.)*
4. Both branches → **perturbed VEGFR3-dependent signalling during cardiac outflow-tract / second-heart-field morphogenesis → altered heart size and looping** (shown by zebrafish *flt4* depletion, rescued by WT but not TOF-variant mRNA) → maldevelopment of the right-ventricular outflow tract, pulmonary valve, and great-vessel/arch structures. *(Demonstrated: zebrafish, PMID: 38713105.)*
5. → **Structural congenital heart defect at birth** (Tetralogy of Fallot; absent pulmonary valve; right aortic arch; other conotruncal/left-sided lesions). *(Demonstrated: human cohorts.)*
6. → After surgical repair, chronic **pulmonary regurgitation → RV volume overload → RV dilation → atrial/ventricular arrhythmia** across adulthood. *(Demonstrated: natural-history cohorts, PMID: 42258347.)*

Contrast with Milroy disease (same gene, different mechanism): Milroy-causing FLT4 missense variants **prevent the receptor from functioning as a homodimer** (dominant-negative), impairing lymphatic rather than cardiac endothelium and producing congenital lymphedema PMID: 33067626, PMID: 38201272.



Pathways/GO/CL terms: VEGF/VEGFR3 signalling (KEGG hsa04370; Reactome VEGF signalling); GO:0001525 angiogenesis, GO:0001946 lymphangiogenesis, GO:0003007 heart morphogenesis, GO:0003151 outflow tract morphogenesis, GO:0034976 response to ER stress, GO:0048010 VEGFR signalling pathway. **Cell types (CL):** endothelial cell (CL:0000115), cardiac/endocardial endothelial cell (CL:0010008), lymphatic endothelial cell (CL:0002138 — relevant to the Milroy branch). **Subcellular (GO CC):** endoplasmic reticulum (GO:0005783), plasma-membrane receptor complex.

7. Anatomical Structures Affected

Primary organ: heart (UBERON:0000948), specifically the **right ventricular outflow tract** (UBERON:0004145), **pulmonary valve** (UBERON:0002146), **interventricular septum** (UBERON:0002094), and **aorta/aortic arch** (UBERON:0000947 / UBERON:0005805). **Body system:** cardiovascular (and, via VEGFR3's canonical role, the lymphovascular system in the allelic Milroy phenotype). **Secondary involvement:** right ventricle (dilation/overload), right atrium (arrhythmia substrate), pulmonary arteries (including MAPCAs in severe TOF variants

PMID: 41957907). **Tissue/cell level:** vascular/endocardial **endothelium** (CL:0000115), cardiac neural-crest/second-heart-field derivatives of the outflow tract. **Lateralization:** right aortic arch reflects a laterality shift; lesions are structural and typically central/midline.

8. Temporal Development

Onset: congenital — the structural defect is present at birth and typically detected prenatally or in the neonatal/infant period. **Progression:** the primary lesion is static once formed, but the **post-repair course is progressive**: chronic pulmonary regurgitation leads to RV volume overload and dilation over years to decades, with rising arrhythmia burden by mid-adulthood (atrial tachyarrhythmia/AF ~20–45% by age 45) PMID: 42258347. **Critical period:** cardiac outflow-tract morphogenesis in early embryogenesis is the window of vulnerability; the intervention window is surgical repair in infancy/early childhood. **Duration:** chronic/lifelong, requiring lifelong ACHD follow-up and periodic reintervention PMID: 42894905.

9. Inheritance and Population

Inheritance: autosomal dominant, **incomplete penetrance**, **variable expressivity**; both de novo and transmitted alleles, with transmitting parents often unaffected or subclinically affected PMID: 39870876, PMID: 32037394, PMID: 40127276. **Epidemiology (phenotype-level):** TOF prevalence ~3/10,000 live births, 5–10% of all CHD PMID: 36277185; FLT4 variants explain ~5% of TOF probands PMID: 30232381 and, with NOTCH1, ~7% of nonsyndromic TOF PMID: 30582441. **Penetrance/anticipation/mosaicism/founder/consanguinity:** incomplete penetrance established; no anticipation (not a repeat-expansion disorder); germline mosaicism not specifically quantified; no founder effect described for CHTD7; consanguinity is not a feature of this dominant entity (note: a rare *recessive* FLT4 lymphedema allele exists PMID: 26091405, separate from CHTD7). **Sex ratio/geography:** no strong sex skew or geographic clustering documented for FLT4-CHD specifically.

10. Diagnostics

Imaging: transthoracic **echocardiography** is the primary diagnostic; cardiac MRI quantifies RV volumes and pulmonary regurgitation; CT/angiography defines pulmonary arteries/MAPCAs/arch anatomy PMID: 41957907. **Electrophysiology:** ECG/Holter for arrhythmia surveillance PMID: 42403553. **Genetic testing:** given locus heterogeneity, **genome or exome sequencing** / **multigene CHD panels** (including FLT4, NOTCH1, KDR) are the recommended approach; genome sequencing is anticipated to become a first-tier test PMID: 32037394. Single-gene FLT4 testing and chromosomal microarray/CNV analysis are useful given the deletion/

CNV spectrum (Finding 5). RNA-sequencing can resolve splice-site variants of uncertain significance PMID: 39870876. **Clinical criteria/differential:** TOF is defined anatomically; differential diagnosis includes 22q11.2 deletion (DiGeorge/syndromic TOF), Alagille (JAG1/NOTCH2), and other conotruncal syndromes. The large **VUS burden** (623/741 FLT4 records) complicates interpretation (Finding 5). **Screening:** fetal echocardiography; cascade genetic testing of relatives given AD inheritance.

11. Outcome / Prognosis

Survival: excellent — **~90% of TOF patients survive >30 years after repair** PMID: 42258347; contemporary adult cohorts report long-term mortality as low as ~3.6% PMID: 42894905, though late-presenting/low-resource settings have higher operative mortality (~11.6%) PMID: 42277928. **Morbidity:** near-universal pulmonary regurgitation → RV volume overload; reintervention (chiefly pulmonary valve replacement) in **~41–52%** PMID: 42894905; atrial and ventricular arrhythmias are major late morbidities and sudden-death risks PMID: 42403553. **Prognostic factors:** anatomic subtype (e.g., TOF with MAPCAs/absent pulmonary valve), age at repair, RV size/function, and arrhythmia burden. **Quality-of-life:** reduced by exercise limitation, reinterventions, and arrhythmia; lifelong ACHD follow-up is required.

12. Treatment

Mainstay — surgery (NCIT: Cardiac Surgical Procedure): complete TOF repair (VSD closure + RVOT relief/transannular patch or homograft), typically in infancy PMID: 42258347, PMID: 42894905. Severe anatomy (MAPCAs, ductal-origin PA) is managed with staged unifocalization/ductal stenting then repair PMID: 41957907. **Late reinterventions:** pulmonary valve replacement — surgical (including minimally invasive left anterior thoracotomy) or transcatheter PMID: 42875919. **Arrhythmia management:** catheter ablation (including pulsed-field for AF), substrate-guided VT ablation, conduction-system pacing, and selective ICD/subcutaneous/leadless devices PMID: 42403553. **Advanced/targeted therapy: none FLT4-specific exists**; VEGF-C gene therapy rescues the lymphatic phenotype in the Chy mouse PMID: 11592985 but is not applicable to the cardiac branch, and **proteostasis modulation** is only a preclinical lead from the ER-aggregation mechanism PMID: 38713105. **Pharmacogenomics:** not established. **Supportive care:** heart-failure and arrhythmia medical therapy, exercise counselling, nutrition support for late-presenting underweight patients PMID: 42277928.

13. Prevention

Primary prevention of the Mendelian defect is not possible, but maternal periconceptional **folic acid** (OR 0.82) and folate/antioxidant-rich diets reduce CHD risk generally PMID: 19952004, PMID: 34392783, and control of maternal pregestational diabetes addresses a strong modifiable risk factor PMID: 32092068. **Secondary prevention:** fetal/neonatal echocardiographic screening and cascade genetic testing of first-degree relatives (given AD inheritance with incomplete penetrance). **Tertiary prevention:** lifelong ACHD surveillance to detect and treat progressive pulmonary regurgitation, RV dilation, and arrhythmia before decompensation PMID: 42894905. **Genetic counselling:** essential — an affected individual has ~50% transmission risk per pregnancy, but penetrance is incomplete so phenotype prediction is limited. Prenatal/preimplantation genetic testing is available for known familial variants.

14. Other Species / Natural Disease

Orthologs: mouse *Flt4* (NCBI Gene 14257), zebrafish *flt4* (NCBI Gene 30375). VEGFR3 is evolutionarily conserved across vertebrates PMID: 33067626. **Natural disease:** no well-characterized spontaneous companion-animal FLT4-CHD counterpart is documented (OMIA entries for this specific gene–CHD association are not established); the comparative value is experimental rather than veterinary. **Conservation:** the VEGF-C/VEGFR3 axis and its role in cardiovascular and lymphovascular development are conserved, underpinning the utility of zebrafish and mouse models.

15. Model Organisms

Zebrafish (cardiac branch, most CHTD7-relevant): *flt4* depletion → reduced heart size and altered looping, rescued by WT human *FLT4* mRNA but not by TOF-variant mRNA; proteostasis inhibition abrogates the defect PMID: 38713105. Zebrafish also reveal Sox18–VegfC genetic interaction in lymphangiogenesis PMID: 23520166. **Mouse:** germline heterozygous Vegfr3-inactivating **Chy** mutant models the **lymphatic (Milroy) branch** (hypoplastic cutaneous lymphatics; rescued by VEGF-C gene therapy) PMID: 11592985; homozygous *Flt4*-null is embryonic lethal with cardiovascular failure PMID: 33067626. **In vitro:** primary human endothelial cells expressing TOF variants recapitulate the ER-aggregation/proteostatic mechanism PMID: 38713105. **Limitation:** no mouse model cleanly reproduces the human TOF cardiac phenotype from heterozygous FLT4 loss; the cardiac phenotype is best captured in zebrafish, while the mouse captures the lymphatic allelic series. **Resources:** ZFIN, MGI, IMPC.

Mechanistic Model / Interpretation

CHTD7 is best understood as a **dosage-sensitive endothelial signalling disorder of cardiac outflow-tract development**. Two complementary lines of quantitative evidence anchor a **haploinsufficiency** model: FLT4 is the second genome-wide-significant nonsyndromic-TOF gene after NOTCH1 (cohort-level enrichment of rare deleterious alleles, $P < 5\times10^{-8}$), and it is among the most loss-of-function-intolerant genes in the human genome (gnomAD pLI = 1.0, LOEUF = 0.249, only 30 of 163 expected LoF alleles observed). Together these establish that a **~50% reduction in VEGFR3 activity is developmentally non-neutral** for the heart.

Layered on top of simple dosage loss is a **variant-class-specific toxic mechanism**: TOF missense alleles cause mutant receptor to aggregate in the perinuclear ER and trigger proteostatic/metabolic stress in endothelial cells — a mechanism demonstrated in primary human endothelial cells and functionally validated in zebrafish, and one that is **mechanistically separable from the dominant-negative, homodimer-blocking missense alleles that cause Milroy lymphedema**. This elegant allelic dichotomy — same gene, two diseases, two mechanisms, two target endothelia (cardiac vs lymphatic) — is the central conceptual insight of this entry and explains why CHTD7 and Milroy disease, though allelic, are phenotypically distinct and rarely co-segregate.

Feature	CHTD7 (FLT4-TOF)	Milroy disease (allelic)
Phenotype	Tetralogy of Fallot / conotruncal CHD	Congenital lymphedema
Target endothelium	Cardiac/vascular	Lymphatic
Dominant variant class	LoF + damaging missense	Missense (kinase-affecting)
Mechanism	Haploinsufficiency + ER aggregation/ proteostatic stress	Dominant-negative (no homodimer)
Best model	Zebrafish *flt4*	Chy mouse
Inheritance	AD, incomplete penetrance	AD

The **incomplete penetrance** that pervades FLT4-CHD (transmission from unaffected/ subclinical parents; 10.1% of a 11,555-proband CHD cohort explained by dominant genes with incomplete penetrance) implies the existence of genetic modifiers and/or environmental

contributors. While general-CHD environmental factors (maternal diabetes risk; folate protection) are well quantified, **no FLT4-specific gene–environment interaction has yet been demonstrated** — a clear knowledge gap.

Evidence Base

PMID	Role in this report
30232381	Core: 5.1% of TOF probands carry FLT4 variants (7 LoF incl. 8-kb deletion); absent pulmonary valve & right arch enrichment
30582441	Core: FLT4 genome-wide-significant, 2nd after NOTCH1, ~7% of TOF
38713105	Core: ER-aggregation/proteostatic mechanism; zebrafish rescue; distinct from Milroy
33067626	VEGFR3 biology; dominant-negative Milroy mechanism; homozygous-null lethality
39870876	FLT4 splice variants cause left- and right-sided CHD; incomplete penetrance; RNA-seq diagnostics
32037394	FLT4 transmitted from subclinical parent; genome sequencing as first-tier test
40127276	60 dominant CHD genes; 10.1% of 11,555 probands; incomplete penetrance, de novo + transmitted
34328347	Reanalysis of 811 TOF probands confirms FLT4 as major contributor
42258347	Prognosis: ~90% >30-yr survival; universal PR/RV overload; AF 20–45% by 45
42894905	Long-term adult cohort: reintervention ~41–52%, PVR most common
36277185	TOF epidemiology (~3/10,000; 5–10% of CHD); nonsyndromic genetics
11592985	Chy mouse models Milroy branch; VEGF-C gene-therapy rescue
32092068	CHD risk: maternal diabetes adjOR 4.04; folate-rich diet protective
19952004	Periconceptional folic acid protective (OR 0.82)
34392783	Mediterranean/antioxidant diet protective against congenital heart defects
27443814	Secondhand smoke associated with septal defects
42403553	Arrhythmia management in adult CHD
42875919	Minimally invasive pulmonary valve replacement after TOF repair
41957907	TOF with MAPCAs/ductal-origin PA anatomy and outcomes
42277928	TOF surgical outcomes in developing-country setting
26091405	Rare recessive FLT4 lymphedema allele (distinct from CHTD7)
22745786	VEGFR3 → PI3K/AKT, MEK/ERK downstream signalling

PMID	Role in this report
38201272	Dysregulation of lymphatic endothelial VEGFR3 signalling
23520166	Sox18–VegfC genetic interaction in zebrafish lymphangiogenesis

Evidence types: human clinical/genetic cohorts (30232381, 30582441, 34328347, 40127276, 39870876, 32037394), in vitro endothelial models (38713105), model organisms (38713105 zebrafish; 11592985 mouse), and epidemiological case-control studies (32092068, 19952004).

Limitations and Knowledge Gaps

- **Serial-number disease label:** CHTD7 lacks dedicated Orphanet/ICD/MeSH codes and recorded synonyms; most clinical literature discusses the TOF phenotype rather than the OMIM entity by name, so phenotype-level and gene-level data had to be bridged.
- **Penetrance not quantified:** incomplete penetrance is firmly established but no numeric penetrance estimate specific to FLT4-CHD is available; modifier genes are unmapped.
- **No FLT4-specific GxE interaction:** environmental risk/protective data (diabetes, folate) are from general-CHD cohorts, not FLT4 carriers.
- **VUS burden:** 623/741 FLT4 ClinVar records are VUS, limiting clinical actionability; functional assays (ER-aggregation readout) are not yet routine.
- **Model gap:** no heterozygous mouse cleanly reproduces the human TOF cardiac phenotype; the mouse captures the lymphatic (Milroy) branch while zebrafish captures the cardiac branch.
- **No targeted therapy:** proteostasis modulation and VEGF-C gene therapy are preclinical/lymphatic-only; no cardiac-directed molecular therapy exists.
- **Epidemiology of the molecular subtype:** prevalence/incidence, sex ratio, and geographic distribution are reported at the TOF phenotype level, not the FLT4-genotype level.

Proposed Follow-up Experiments / Actions

1. **Penetrance estimation:** assemble FLT4-variant pedigrees with genotyped relatives to derive an age-/sex-adjusted penetrance estimate and test for modifier loci (including NOTCH1/KDR second hits).
2. **Functional VUS triage:** develop a scalable ER-aggregation/proteostatic-stress assay in primary human endothelial cells (per PMID: 38713105) to reclassify the large FLT4 VUS pool.
3. **Mechanism-anchored therapeutics:** test proteostasis modulators / ER-stress-pathway inhibitors in the zebrafish *flt4*-TOF-variant model to validate the preclinical lead.
4. **Genotype–phenotype mapping:** correlate FLT4 variant class (null vs ER-aggregating missense) with anatomic subtype (absent pulmonary valve, right aortic arch, MAPCAs) and long-term reintervention/arrhythmia outcomes.
5. **Gene–environment interaction:** prospectively test whether maternal diabetes/folate status modifies penetrance specifically in FLT4-variant pregnancies.
6. **Cardiac mouse model:** engineer conditional/endothelial-specific humanized FLT4-TOF-missense knock-in mice to close the model-organism gap for the cardiac phenotype.

Report compiled from 8 confirmed findings across 29 reviewed papers over 5 iterations. Causal gene: FLT4 (VEGFR3). OMIM 618780; MONDO:0032913.
